Minding the Retrieval Gap in RAG: *An Oracle vs. Embedding RAG Analysis on Multilingual Regional QA*

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Models Tested

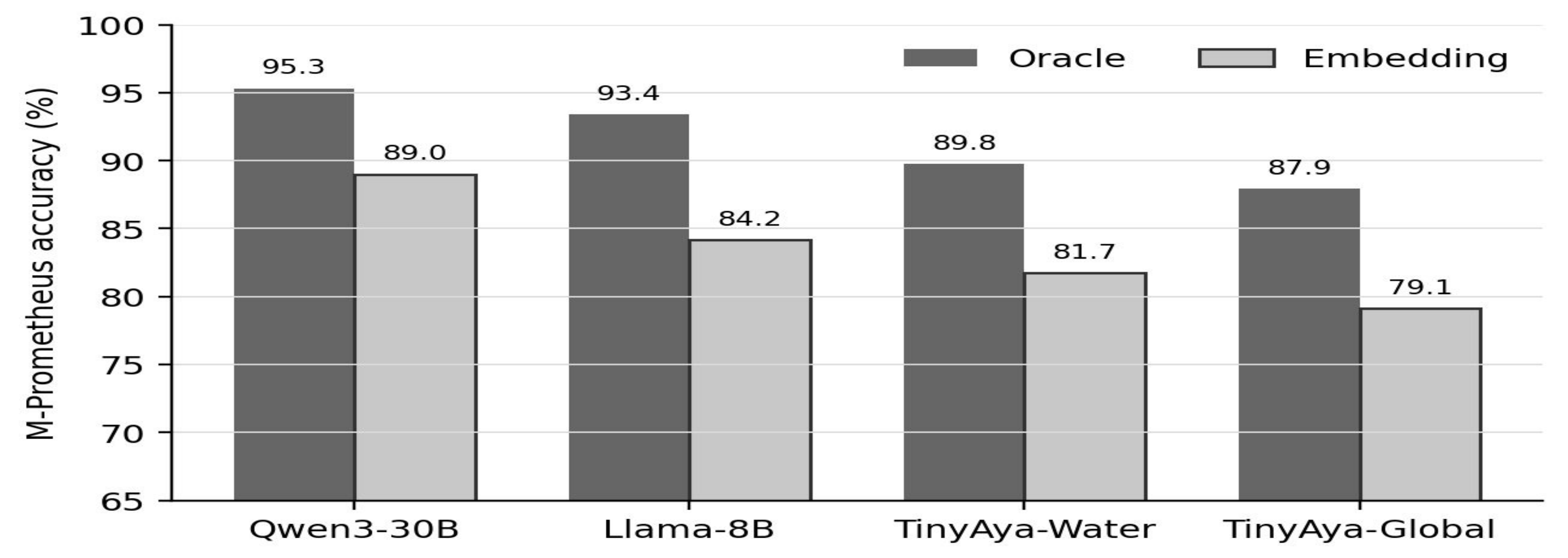
Tiny Aya Global & Water (**3.35B** multilingual) · **Llama-3.1-8B** (open-weight baseline) · **Qwen3-30B-A3B** (strongest that I could run reliably)

Tiny Aya Water is specialised for **Europe** and Asia-Pacific; Global is the broad, multilingual version.

Four Research Questions

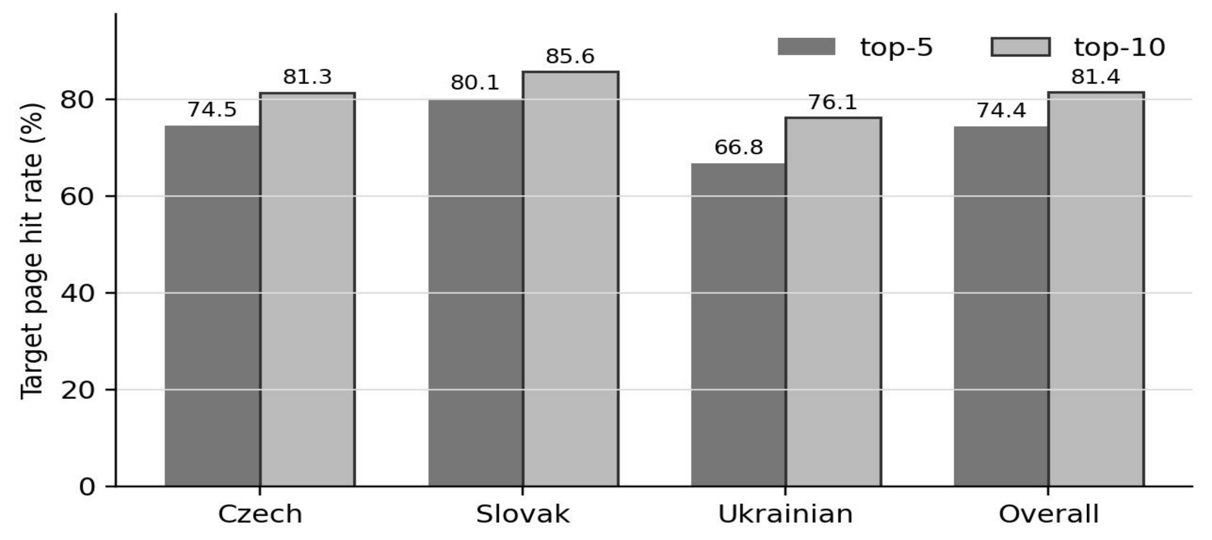
- ***Q1:*** *How large is the gap between exact-source Oracle RAG and realistic Embedding RAG?*
- ***Q2:*** *Does increasing retrieval depth from top-5 to top-10 help, or does it distract?*
- ***Q3:*** *Which open-weight generator gives the best quality under runtime and memory constraints?*
- ***Q4:*** *Do semantic LLM judging and lexical chrF evaluation lead to the same conclusions?*

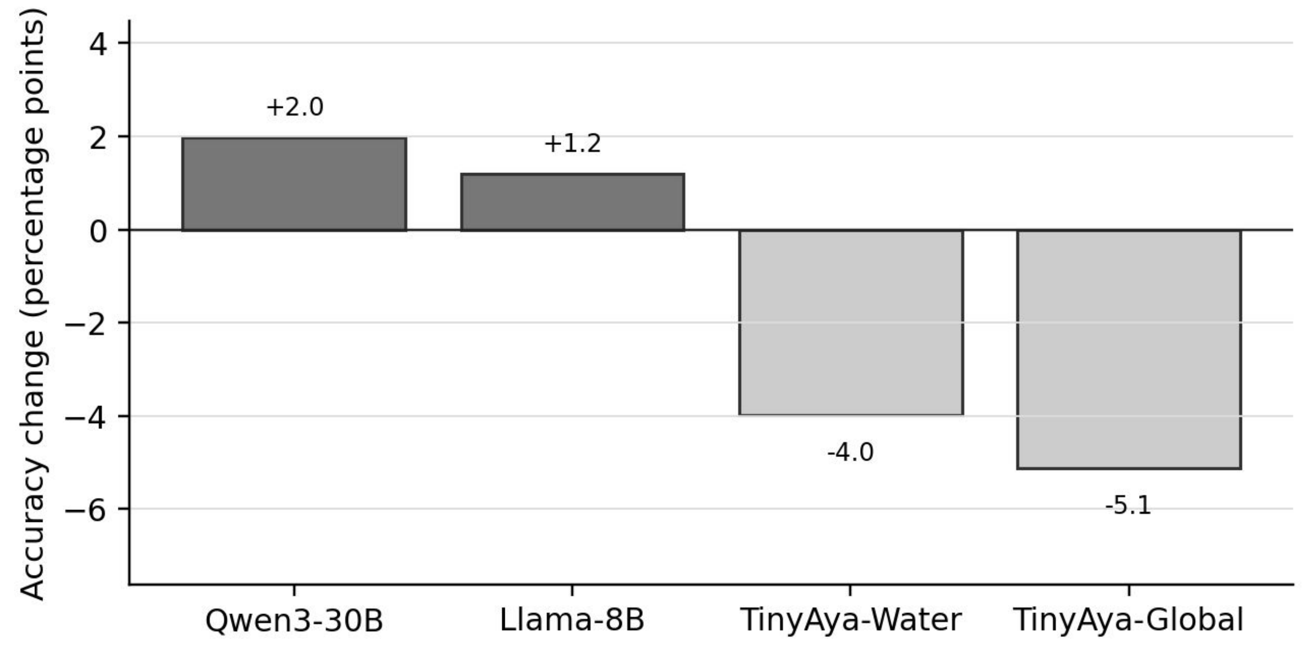
Q1: Oracle vs. Embedding RAG Gap



Finding: For Qwen3-30B, accuracy drops from **0.953** (Oracle) to **0.890** (Embedding). ***Retrieval remains the main bottleneck.***

Q2: Does Top-10 *k* Help or Distract?





Finding: *Target page Hit rate* rises 74.4% → 81.4%, but Qwen/Llama improve while Tiny Aya models worsen with top-10.

Takeaway: Coverage-vs-distraction tradeoff: larger models *use extra evidence*, while smaller models get *distracted* by it.

Q3: Best Generator Under Runtime & Memory Constraints

System (Embedding)	Setting	Acc.	Correct	Runtime	Peak GPU Mem	+Correct Answer
TinyAya-Water	top-5	0.8175	1151 / 1408	0:19:29	7.7 GiB / 1 GPU	—
TinyAya-Global	top-5	0.7912	1114 / 1408	0:19:42	7.7 GiB / 1 GPU	-37
Llama-8B	top-10	0.8416	1185 / 1408	1:11:56	16.4 GiB / 1 GPU	+34
Qwen3-30B	top-10	0.8899	1253 / 1408	2:54:08	60.7 GiB / A100	+68

Finding: Qwen best overall (0.890 acc, 51.09 chrF). **Tiny Aya Water** ~2.4 pts behind **Llama**, 3.5x faster, <50% GPU memory.

Q4: chrF vs. M-Prometheus Evaluation Agreement

System	Condition	k	Generator	chrF	M-Prom.	Judge rank / chrF
Oracle Qwen3-30B	Oracle	–	Qwen3-30B	56.52	0.953	1 / 1
Oracle Llama-8B	Oracle	–	Llama-8B	45.89	0.934	2 / 4
Oracle TinyAya-Water	Oracle	–	TinyAya-Water	39.89	0.898	3 / 7
Qwen3-30B top-10	Embedding	10	Qwen3-30B	49.42	0.890	4 / 3
Oracle TinyAya-Global	Oracle	–	TinyAya-Global	39.87	0.879	5 / 8
Qwen3-30B top-5	Embedding	5	Qwen3-30B	49.95	0.870	6 / 2
Llama-8B top-10	Embedding	10	Llama-8B	40.51	0.842	7 / 6
Llama-8B top-5	Embedding	5	Llama-8B	40.52	0.830	8 / 5
TinyAya-Water top-5	Embedding	5	TinyAya-Water	37.10	0.818	9 / 10
TinyAya-Global top-5	Embedding	5	TinyAya-Global	37.29	0.791	10 / 9
TinyAya-Water top-10	Embedding	10	TinyAya-Water	36.56	0.778	11 / 12
TinyAya-Global top-10	Embedding	10	TinyAya-Global	37.07	0.740	12 / 11

Finding: Both agree **Qwen** is strongest, but rank middle systems *differently*. **chrF** rewards character overlap; **M-Prometheus** judges semantic correctness.

Final Takeaway

Retrieval remains the biggest gap: improving RAG is not just about larger generators or more context, but an engineering problem of finding, ranking, and filtering the right evidence.